

Post-Quantum Cryptography Migration on V2X Certificate using KpqC Algorithms*

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Abstract—Connected vehicles utilizing Vehicle-to-Everything (V2X) communication enhance road safety and transportation efficiency, supporting cooperative autonomous driving through real-time interactions. However, increased connectivity raises cyber-attack risks, endangering driver and pedestrian safety. This highlights the urgent need to integrate Post-Quantum Cryptography (PQC) into vehicular communications [6]. In this paper, we implement the Korean PQC digital signature algorithm HAETAE for V2X environments and compare its performance with the NIST PQC signature scheme, ML-DSA which is derived from CRYSTALS-DILITHIUM, and traditional signatures, RSA, and ECC under TLS 1.3 environments. Results indicate that PQC algorithms introduce substantial overhead, whereas traditional algorithms produce smaller certificates. Specifically, HAETAE provides more efficient certificates than ML-DSA [3], minimizing latency impacts in TLS operations. These findings inform the critical balance between enhanced security and certificate size, guiding future post-quantum TLS designs.

Index Terms—Post-Quantum Cryptography, TLS, V2X, KpqC, signature, certificates.

I. INTRODUCTION

The rapid advancement of quantum computing necessitates the adoption of quantum-resistant cryptographic schemes. Traditional public-key cryptographic algorithms, including RSA and ECC, have long provided secure communications in TLS protocols with X.509 certificates for validating public keys [4]. However, these algorithms are now vulnerable to quantum attacks, driving a transition towards post-quantum cryptographic (PQC) methods. This study investigates the certificate size overhead associated with two leading PQC signature algorithms—HAETAE and ML-DSA—compared to conventional algorithms RSA and ECC.

Our experimental framework is established on a Linux operating system (Ubuntu 22.04) and utilizes the `liboqs` with KPQC signature algorithm, HAETAE. The certificates are generated following the X.509v3 standard and are encoded in DER (Distinguished Encoding Rules) format to ensure platform independence and compliance with industry standards.

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We employ SHA-256 for all signature operations to maintain consistency across our measurements.

The simulation results provide concrete numerical insights into the trade-offs inherent in these cryptographic systems. For instance, HAETAE-based certificates exhibit a clear correlation between security level and certificate size, with sizes ranging from 2813 bytes at the HAETAE-120 level to 5375 bytes at the HAETAE-260 level. In contrast, traditional algorithms such as RSA and ECDSA produce much smaller certificates, with RSA at approximately 891 bytes and ECDSA ranging from 475 to 630 bytes. Moreover, while ML-DSA results in even larger certificate sizes, HAETAE proves to be more size-efficient among post-quantum alternatives.

This comprehensive analysis underscores the implications of increased size on certificates due to new PQC standard algorithms. It sets the stage for future optimization of TLS performance in the emerging post-quantum era. The findings provided a foundation for subsequent research incorporating additional signature algorithms and Key Encapsulation Mechanisms (KEMs), evaluating their impacts on security and performance in practical TLS environments.

II. PRELIMINARIES

PQC comprises cryptographic algorithms explicitly designed to resist quantum computer attacks. Recognizing the urgency of quantum-resistant cryptography, the National Institute of Standards and Technology (NIST) initiated a PQC standardization project in 2017. By 2022, NIST selected the first four standard algorithms and subsequently included HQC as an additional standard Key Encapsulation Mechanism (KEM) algorithm in 2025. Similarly, Korea independently selected four algorithms conforming to national representative PQC schemes in 2025.

The primary PQC signature algorithm selected by NIST is CRYSTALS-DILITHIUM, while Korea's primary PQC signature algorithm is HAETAE. Both algorithms fall under lattice-based cryptography [3]. HAETAE, in particular, has been recognized by the Korean PQC community (KpqC) for its high security. Although HAETAE demonstrates improved parameter sizes compared to Dilithium at equivalent security levels, its computational performance is slower, necessitating further optimization research.

TABLE I: Parameters of HAETAE Algorithm. Security level (SL), signature and key sizes (bytes). Sum column represents the sum of public key and signature sizes.

Scheme	SL	Public Key	Signature	Sum	Secret Key
HAETAE-120	2	992	1,474	2,466	1,408
HAETAE-180	3	1,472	2,349	3,821	2,112
HAETAE-260	5	2,080	2,948	5,028	2,752

HAETAE is a lattice-based digital signature scheme that improves Dilithium by reducing the signature and key size by up to 39%. It is designed based on the module structure. The main feature of HAETAE is that the probability distribution for rejection sampling is sampled in the hyperball uniform distribution by using a binary distribution and changing the rejection target distribution. HAETAE introduces a new rejection procedure to minimize the secret key size when signing, which has been improved to perform rejection sampling more efficiently in the signature process, resulting in a smaller signature size. It applies a technique that minimizes the impact of verification key truncation by correcting the signature key. The signature key increases somewhat in this process, but an efficient correction method is applied to minimize the increase in signature key size. HAETAE proposes three parameter sets for each NIST security level: 2, 3, and 5, respectively, HAETAE-120, HAETAE-180, and HAETAE-260 [3].

Table I shows the parameter sizes of HAETAE. These parameters are smaller than the public and secret keys, and the signature size corresponding to each parameter of ML-DSA—i.e., the standardized form of CRYSTALS-DILITHIUM was adopted as the NIST standard signature algorithm. When generating a certificate, HAETAE maintains a relatively small size and contributes to reducing overhead in TLS handshake.

A digital certificate is an electronic document used to identify an entity, establish the owner’s identity, and provide the owner (or subject) with a public key. Specifically, X.509 certificate generates a certificate in the data section, including X.509 version, serial number, algorithm information, Issuer distinguished name, expiration date, Subject distinguished name, and Subject public key information. In addition, the signature section contains the encryption algorithm, the algorithm used by the issuer to generate the digital signature and the digital signatures of the CA. This value is the hash of all information in the certificate, encrypted with the private key [4].

III. PROPOSED METHOD

This study analyzes the impact of post-quantum algorithms, such as HAETAE and ML-DSA, on the certificate of the TLS protocol for V2X applications. We compare the performance of PQC schemes against traditional public-key algorithms RSA and ECC.

Our experimental framework runs on Ubuntu 22.04. We utilize the `liboqs` library, which includes reference implementations of NIST PQC standard algorithms and ML-DSA

schemes (ML-DSA-44, ML-DSA-55, ML-DSA-66). Furthermore, we implemented reference versions of the HAETAE scheme (HAETAE-120, HAETAE-180, HAETAE-260) within a modified `liboqs` library, allowing these schemes to serve as X.509 certificate signature algorithms.

We incorporated additional signature algorithms to enable a broader comparative analysis. HAETAE was compared to traditional RSA, ECC (secp256r1), and PQC standards such as ML-DSA. All signature operations employed SHA-256 to maintain uniformity in security parameters and performance measurements. Certificates were encoded consistently using Distinguished Encoding Rules (DER), facilitating the interchangeability of signature algorithms and hash functions.

A. Study Design and Implementation Details

The core idea behind our method is to analyze the certificate size after implementing a standardized X.509 certificate generation process. We used the X509 set version to specify that certificates conform to the X.509v3 standard. Each certificate is constructed by setting several key components. Subject and Issuer Information, Public Key Inclusion, Validity Period, Serial Number, and Signature [4].

Our method analyzes certificate sizes produced by the standardized X.509v3 certificate generation process. Certificates are constructed using key components, including subject and issuer information, public key, validity period, serial number, and signature [4]. Certificates are uniformly encoded using DER to ensure platform independence and compliance with the X.509 standard. This consistent encoding method enables seamless substitution of signature algorithms and hash functions across experiments.

The design specifically addresses delay-sensitive requirements in applications such as V2X communications [6]. Our framework evaluates the effect of traditional (RSA, ECC) and post-quantum signature algorithms (ML-DSA and HAETAE) on certificate size, impacting the number of Ethernet packets required for transmission. This modular approach facilitates comparative analysis, highlighting implications for packet fragmentation and transmission delay in V2X scenarios.

IV. NUMERICAL RESULTS

The Table II details the certificate generation outputs for each algorithm. The core idea behind our implementation is to analyze the certificate size resulting from a standardized X.509 certificate generation process. As detailed in Table I, the certificate sizes for traditional public-key algorithms are significantly smaller than those for PQC algorithms. For instance, RSA-SHA256 and ECDSA-SHA256 produce certificates of 891 and 495 bytes, respectively, sizes that easily fit within a single Ethernet transmission packet (with a standard MTU of 1,500 bytes). In contrast, PQC schemes yield substantially larger certificates. Specifically, the ML-DSA variants have certificate sizes of 4097, 5626, and 7584 bytes for ML-DSA-44, ML-DSA-55, and ML-DSA-66, respectively, while the HAETAE implementations yield sizes of 2813, 4168, and 5375

TABLE II: Each algorithms of NIST Security level and the size of certificate, public key, secret key and signature (bytes)

Algorithm	Key, Signature Size	NIST Security level	Certificate Size
RSA-SHA256	pk 270~300	/	891
	sk 1190		
	sig 256		
ECDSA-SHA256	pk 65	5	495
	sk 32		
	sig 64		
ML-DSA-44	pk 1312	2	4097
	sk 2560		
	sig 2420		
ML-DSA-65	pk 1952	3	5626
	sk 4032		
	sig 3293		
ML-DSA-87	pk 2592	5	7584
	sk 4896		
	sig 4595		
HAETAE-120	pk 992	2	2813
	sk 1408		
	sig 1474		
HAETAE-180	pk 1472	3	4168
	sk 2112		
	sig 2349		
HAETAE-260	pk 2080	5	5375
	sk 2752		
	sig 2948		

*All parameter sizes for RSA-SHA256 are based on DER encoding.

bytes for HAETAE-120, HAETAE-180, and HAETAE-260, respectively.

A. Results Analysis

Given that an Ethernet packet can carry a maximum of 1500 bytes (excluding header overhead), certificates exceeding this size must be fragmented across multiple packets. This fragmentation naturally increases the number of packets required for a complete transmission [7]. While both ML-DSA and HAETAE certificates exceed the 1-MTU limit, Fig. 1 shows that HAETAE certificates remain consistently smaller than their ML-DSA counterparts for a given security level. For example, at NIST security levels (SL) corresponding to parameters 2, 3, and 5, the HAETAE-based certificates (HAETAE-120, HAETAE-180, HAETAE-260) require fewer packets than those generated by ML-DSA.

This reduction in required transmission packets is of particular importance in V2X (Vehicle-to-Everything) communications, where low latency is critical. In V2X scenarios, any increase in transmission delay—such as that introduced by packet fragmentation and reassembly—can negatively impact system performance and safety [5]. Therefore, the smaller certificate sizes offered by HAETAE not only lower the overhead in TLS handshakes, but also present a significant advantage for delay-sensitive V2X applications, potentially leading to improved reliability and responsiveness in vehicular communication networks.

V. CONCLUSION

In this study, we conducted a comparative analysis of certificate sizes generated by various signature algorithms, including

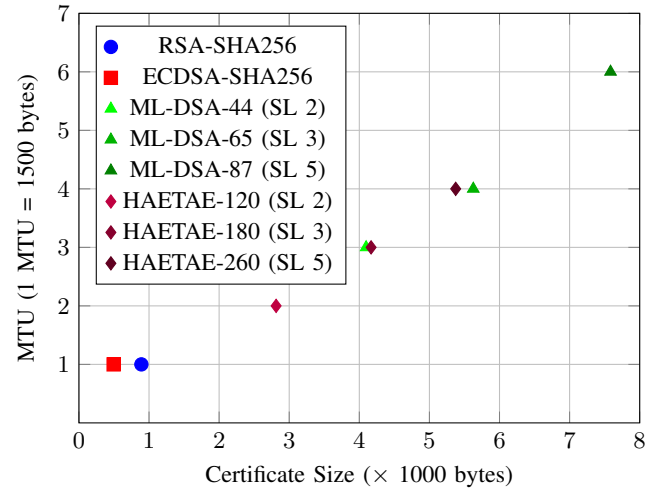


Fig. 1: Certificate size vs. Required MTUs for each algorithms.

traditional schemes (RSA and ECC) and PQC algorithms (ML-DSA and HAETAE). Although traditional algorithms produce certificates comfortably fitting within a single Ethernet packet (1 MTU = 1500 bytes), PQC schemes inherently generate larger certificates due to longer key lengths and additional security parameters. Our experimental results indicate that although both ML-DSA and HAETAE certificates exceed the 1-MTU limit, HAETAE certificates consistently remain smaller than their ML-DSA counterparts for a given security level.

This reduction in certificate size leads to fewer required transmission packets, which is a significant advantage in delay-critical applications such as V2X communications. In such scenarios, minimizing packet fragmentation and reassembly delays can substantially enhance system performance and safety [5]. Therefore, our findings suggest that, within the post-quantum category, HAETAE offers a more favorable trade-off between security and transmission efficiency than ML-DSA.

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